FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 — SUB-STRAND 1.2: INDICES	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

Purpose of This Document

This Final Explanation document is designed to help you show your complete understanding of indices and logarithms by pulling together everything you explored across all 8 lessons. Think of it as your opportunity to tell the full story — from the Maize Doubling Puzzle in Kitale, all the way to the laws of logarithms and their role in Kenyan farming, finance, engineering, and science.

The Driving Question You Are Answering

How can writing numbers as powers and logarithms help us make sense of — and calculate with — the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science?

What Is a Final Explanation?

A Final Explanation is NOT a list of rules and formulas. It is a carefully reasoned, evidence-based argument that shows:

- **What** the key concepts of indices and logarithms are
- **Why** those concepts work the way they do (the reasoning behind the rules)
- **How** you know (specific evidence from your investigations and activities)
- **Where** these ideas connect to real life in Kenya and beyond

---

How to Write Your Final Explanation — Step by Step

1. **Review your evidence first.** Before you write a single sentence, go back through your notes, your Summary Table, your initial prediction models, and every investigation you completed. Look for patterns, surprises, and moments where your thinking changed.

2. **Use the four sections below as your structure.** Each section has a guiding prompt and a worked example to show you the level of depth expected. Do not copy the example — use it as a quality benchmark for your own writing.

3. **Cite specific activities and observations.** Vague statements score poorly. Strong explanations sound like:
  - *"When we investigated the Maize Doubling Puzzle, I noticed that Season 8 gives $2^8 = 256$ seeds. Multiplying $2^8 \times 2^4$ by writing out all the 2s showed me that we were simply combining 8 twos and 4 twos into 12 twos — that is why $a^m \times a^n = a^{m+n}$."*
  - *"In the compound interest investigation using a Kericho tea cooperative's loan, I found that Ksh 10,000 growing at 8% per year for 10 years equals $10{,}000 \times (1.08)^{10}$. Without logarithms it was almost impossible to find how many years it takes to double — but once I applied log both sides, the answer appeared in two steps."*

4. **Explain the 'why', not just the 'what'.** Do not simply state a law. Unpack the reasoning. Show what would happen if the law were wrong. Connect it back to the phenomenon.

5. **Use correct mathematical notation.** Write indices as superscripts (e.g. 2^8, 10^{-3}, a^m). Write logarithms in full (e.g. $\log_{10} 1000 = 3$, $\log_2 256 = 8$). Show working where it strengthens your explanation.

6. **Write in complete sentences and paragraphs.** Bullet points may be used inside a paragraph to list examples, but your explanation must flow as connected prose.

Sentence Starters (use these if you feel stuck)
- *"The Maize Doubling Puzzle shows that… because…"*
- *"I know that [law/rule] is true because in [specific activity] I observed that…"*
- *"Before this unit I thought… but after [investigation] I now understand that…"*
- *"This connects to real life in Kenya when… because…"*
- *"The relationship between indices and logarithms can be explained as… which means that…"*

Section 1 — The Problem With Big and Small Numbers: Why Indices Exist

Return to the Maize Doubling Puzzle. A smallholder farmer in Kitale starts with 1 maize seed. After 8 seasons of doubling, there are 256 seeds. After 20 seasons, there are more than	The Maize Doubling Puzzle reveals a fundamental problem with ordinary number notation. After Season 8, the farmer has 256 seeds — a manageable number. But by Season 20, the harvest has grown to 1,048,576 seeds, and by Season 30 it exceeds one billion. Writing, comparing, or multiplying these numbers in full becomes slow and error-prone. Scientists and engineers face exactly the same challenge with quantities like the distance from Nairobi to the sun (approximately 150,000,000,000 metres) or the size of a bacterium (0.000001 metres).

1,000,000 seeds. **(a) Explain, using the puzzle as your anchor, why ordinary whole-number notation becomes impractical for very large (or very small) quantities.** **(b) Show how index notation solves this problem. Use at least TWO examples from the puzzle — one large (Season 20 or beyond) and one that involves a fractional or negative index — to illustrate your explanation.** **(c) State and explain the meaning of the zero index law ($a^0 = 1$) and the negative index law ($a^{-n} = 1/a^n$). Why do these laws make sense? What would it mean for the farmer's seed count if n = 0 or if n were negative?** **Remember: Do not just state the laws. Explain WHY they are true using the pattern of repeated multiplication and division.**	Index notation solves this by recording HOW MANY TIMES a base is multiplied by itself, rather than writing out the full result. The 256 seeds after Season 8 become 2^8 — a compact expression that immediately tells us the base (2, the doubling factor) and the number of doublings (8 seasons). Season 20 becomes 2^{20}, which is far easier to write, compare, and manipulate than 1,048,576. The zero index law states that $a^0 = 1$ for any non-zero base a. This makes sense when we follow the pattern: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$. Each step divides by 2. Continuing: $2^0 = 2 \div 2 = 1$. For the farmer, Season 0 means no doubling has yet occurred — there is exactly 1 seed. That is consistent with $2^0 = 1$. The negative index law states that $a^{-n} = 1/a^n$. Following the same pattern, $2^{-1} = 1/2$, $2^{-2} = 1/4$, $2^{-3} = 1/8$. In the farmer's context, a negative season could represent going backwards in time — asking 'how many seeds existed one doubling-cycle before Season 0?' The answer is half a seed, or $1/2 = 2^{-1}$. Negative indices therefore represent repeated division, which is why they produce fractions. This is not a separate rule to memorise — it is the same doubling pattern, extended in the opposite direction.

Section 2 — The Laws of Indices: Rules That Are Not Arbitrary	
The laws of indices are not rules handed down from a textbook. They are consequences of what multiplication actually means. **(a) State ALL the laws of indices you investigated during this unit (product law, quotient law, power of a power, power of a product, power of a quotient, zero index, negative index,**	The Product Law states: $a^m \times a^n = a^{m+n}$. To understand WHY, expand both powers into factors. $2^3 \times 2^4$ means $(2 \times 2 \times 2) \times (2 \times 2 \times 2 \times 2)$. Counting all the 2s gives seven 2s multiplied together: 2^7. We did not invent a rule — we simply counted. The exponents add because we are counting repeated factors, and the two groups of factors join into one group. The Power of a Power Law states: $(a^m)^n = a^{mn}$. Consider $(2^3)^4$. This means $2^3 \times 2^3 \times 2^3 \times 2^3$ — four copies of 2^3. By the Product Law, we add the exponents four times: 3 + 3 + 3 + 3 = 12. So $(2^3)^4 = 2^{12} = 4096$. Multiplying the exponents ($3 \times 4 = 12$) is simply a shortcut for repeated addition of the inner exponent. I initially found the fractional index law ($a^{(1/n)} = \sqrt[n]{a}$) confusing, because I could not picture what it means to multiply a number by itself 'half a time'. The investigation that changed my thinking was applying the Power of a Power Law: $(a^{(1/2)})^2 = a^{(1/2 \times 2)} = a^1 = a$. So $a^{(1/2)}$ is the number that, when squared, gives a — which is exactly the definition of the square root. The law did not

fractional index). **(b) For the Product Law ($a^m \times a^n = a^{m+n}$) and the Power of a Power Law ($(a^m)^n = a^{mn}$), give a detailed explanation of WHY the law works. Use an expansion into factors to show your reasoning — do not just state the rule.** **(c) Choose ONE law you initially found confusing or surprising. Describe what confused you, what evidence or investigation changed your thinking, and how you now understand it.** **(d) Demonstrate how at least TWO of the laws work together to simplify a complex expression involving Kenyan context numbers (e.g. powers of 2 for seed counts, powers of 10 for shilling amounts, or surds from a geometry problem).**	introduce a new idea; it showed that square roots and cube roots are simply index notation in disguise. Combined application: A Mombasa import company records a cargo mass as $3^2 \times 3^5 \div 3^4$ kilograms. Using the Product Law first: $3^2 \times 3^5 = 3^7$. Then using the Quotient Law: $3^7 \div 3^4 = 3^3 = 27$ kg. Two laws, applied in sequence, reduce a complex expression to a simple result — exactly the kind of efficient calculation that makes index notation valuable.

Section 3 — Logarithms: The Inverse Operation That Unlocks the Exponent	
Logarithms answer a question that indices alone cannot easily answer: 'The result is this number — what was the exponent?' **(a) Define a logarithm precisely. Explain the relationship log_a(x) = n $\Longleftrightarrow$ $a^n = x$ in plain language, using the Maize Doubling Puzzle as your context (e.g. 'After how many seasons will the farmer have exactly 1,024 seeds?').** **(b) State and explain the three main laws of logarithms (Product Law, Quotient Law, Power Law). For each, show**	A logarithm answers the question: 'To what power must I raise the base to get this number?' Formally, log_a(x) = n means exactly the same thing as $a^n = x$ — they are two ways of writing the same relationship. In the Maize Doubling Puzzle, we know the farmer starts with 1 seed and doubles every season. If we want to know after how many seasons there will be 1,024 seeds, we are asking: $2^n = 1{,}024$, so n = ? Writing this as a logarithm gives $n = \log_2(1{,}024) = 10$. The farmer needs 10 seasons. The logarithm extracts the exponent. The three laws of logarithms follow directly from the laws of indices: – **Product Law**: log_a(mn) = log_a(m) + log_a(n). This mirrors the index Product Law $a^m \times a^n = a^{m+n}$. If $m = a^p$ and $n = a^g$, then $mn = a^{p+g}$, so log_a(mn) = p + q = log_a(m) + log_a(n). Multiplying numbers becomes adding their logarithms. – **Quotient Law**: log_a(m/n) = log_a(m) − log_a(n). This mirrors the Quotient Law of indices. Dividing numbers becomes subtracting their logarithms. – **Power Law**: log_a(m^k) = k·log_a(m). This mirrors $(a^m)^n = a^{mn}$. A power becomes a coefficient — this is particularly powerful for solving equations where the unknown is an exponent. The change-of-base formula states: log_a(x) = $\log_{10}(x) \div \log_{10}(a)$. This is necessary because most calculators — including those used in Kenyan national examinations — only have a base-10 (log) button and a natural log (ln) button. If I need $\log_2(1024)$, I calculate $\log_{10}(1024) \div \log_{10}(2) = 3.0103 \div 0.3010 = 10$. The formula is derived from the Power Law and allows any logarithm to be

how it corresponds to — and is derived from — a matching law of indices. **(c) Explain the change-of-base formula and show when and why it is useful (e.g. when using a calculator that only has $\log_{10}$ or ln).** **(d) Describe a real Kenyan context where you used logarithms to solve a problem during this unit. What was the question, what equation did you set up, and what did the logarithm allow you to find?**	evaluated using available tools. During our compound interest investigation, I used logarithms to solve a real finance problem: a Nairobi SACCO offers 12% compound interest per year. A member deposits Ksh 5,000. The question was: after how many years will the account hold Ksh 20,000? Setting up the equation: $5000 \times (1.12)^n = 20{,}000$, so $(1.12)^n = 4$. Taking $\log_{10}$ of both sides: $n \times \log_{10}(1.12) = \log_{10}(4)$, giving $n = \log_{10}(4) \div \log_{10}(1.12) = 0.6021 \div 0.0492 \approx 12.2$ years. Without logarithms, finding n would require trial and error with a calculator for a long time. The Power Law of logarithms brought the unknown exponent down to where it could be solved directly.

Section 4 — Indices and Logarithms Across Kenyan Contexts	
Throughout this unit you investigated how indices and logarithms appear in farming, finance, engineering, and science. This section asks you to demonstrate those connections. **(a) Choose TWO different real-world Kenyan contexts from the list below and, for each, explain:** **- What quantity is being measured or modelled** **- Why ordinary linear notation is insufficient** **- How indices or logarithms make the quantity manageable** **- What a specific numerical result means in plain language** **Contexts to choose from: (i) Maize seed doubling / crop yield projections for a Kitale farmer; (ii) Compound interest for a Kisumu fishing**	**Context 1 — Tea Farming Finance in Kericho (Compound Interest):** A Kericho tea cooperative borrows Ksh 100,000 from a bank at 10% compound interest per year. The amount owed after n years is $A = 100{,}000 \times (1.10)^n$. After 10 years: $A = 100{,}000 \times (1.10)^{10} \approx$ Ksh 259,374. The index notation $(1.10)^{10}$ compresses what would otherwise require ten successive multiplications into a single expression. If the cooperative wants to know when the debt will double to Ksh 200,000, logarithms give the answer directly: $(1.10)^n = 2$, so $n = \log_{10}(2) \div \log_{10}(1.10) \approx 7.3$ years. Without logarithms, the cooperative's treasurer would have to compute each year's balance by hand until reaching Ksh 200,000 — a slow, error-prone process. **Context 2 — pH of Kenyan Tea and Lake Bogoria:** The pH scale measures the concentration of hydrogen ions in a solution. $pH = -\log_{10}[H^+]$, where $[H^+]$ is measured in moles per litre. Kenyan black tea has a pH of about 6, meaning $[H^+] = 10^{-6}$ mol/L. Lake Bogoria, a soda lake in the Rift Valley, has a pH of about 10.5, meaning $[H^+] = 10^{-10.5}$ mol/L $\approx 3.16 \times 10^{-11}$ mol/L. The actual concentration is an incredibly small decimal — impossible to compare mentally. The logarithmic pH scale compresses this range into a 0–14 scale where each unit represents a tenfold change in acidity. Without logarithms, a chemist testing water safety for Rift Valley communities would struggle to communicate the enormous difference between safe and dangerous water acidity. The deep reason professionals use indices and logarithms is that multiplication — the dominant operation in growth, decay, and scaling — becomes addition when expressed in logarithms. Addition is cognitively simpler and computationally faster. The laws of indices and logarithms are not shortcuts invented for convenience; they are genuine properties of how multiplication works, expressed in a compact language. Just as a Swahili speaker can communicate in fewer words using a rich vocabulary, a mathematician can handle enormous quantities using the vocabulary of powers and logarithms. In Lesson 1, I predicted that the farmer would need a very large notebook but I did not see why a different type of notation would help — I thought it was just about writing numbers shorter. I now understand that the real power is not cosmetic. Index notation preserves the structure of the calculation (base and exponent), and logarithms convert that structure into arithmetic that humans can perform quickly and exactly. The most surprising discovery was that the laws of logarithms are not separate rules — they are simply the laws of indices, translated into a new language.

cooperative's loan; (iii) Population growth in Nairobi between census years; (iv) The pH scale and acidity of Kenyan tea or Lake Bogoria's alkaline water; (v) Earthquake magnitude on the Richter scale for East Africa's Rift Valley region; (vi) Bacterial growth in a Mombasa food safety laboratory. **(b) Explain, in your own words, why scientists, engineers, and bankers — as described in the driving question — 'always seem to compress huge numbers into short expressions'. What is the deep mathematical reason, and what practical advantages does it give?** **(c) Revisit your initial prediction from Lesson 1. How has your understanding changed? What was the most surprising thing you learned?**	

Section 5 — Your Revised Model and the Driving Question Answer	
In Lesson 1, you drew or wrote an initial model to answer the driving question. Now, at the end of the unit, you will construct your final, revised model and write your complete answer. **(a) Draw or describe your REVISED MODEL. Your model must show:** **- The relationship between a number in ordinary form, in**	**Revised Model Description:** At the centre of my model is a triangle connecting three representations of the same idea. At one vertex: the ordinary number (e.g. 1,048,576). At the second vertex: the index form (2^{20}). At the third vertex: the logarithm ($\log_2(1{,}048{,}576) = 20$). Arrows labelled 'evaluate' and 'take log' connect the vertices, showing that these are not different mathematical objects — they are three ways of describing the same quantity. Radiating outward from the index vertex are the four laws (Product, Quotient, Power of a Power, Fractional Index), each with a short example. Radiating from the logarithm vertex are the three logarithm laws, each labelled with its corresponding index law. One large arrow at the bottom points to a Kenyan context panel showing: Kitale maize farm (doubling, 2^n), Kericho tea SACCO (compound interest, $(1.10)^n$), and Rift Valley pH testing (10^{-pH}). **Final Answer to the Driving Question:** Writing numbers as powers and logarithms solves a genuine, practical problem that the Maize Doubling Puzzle made vivid. When a Kitale farmer's seed count grows from 1 to over 1,000,000 in just 20 seasons, writing the full number becomes impractical, comparing two such numbers becomes confusing, and multiplying them by hand becomes nearly impossible. Index notation — writing 2^{20} instead of 1,048,576 — compresses the number while preserving the two most

index form, and as a logarithm (e.g. 1024 = 2^{10} ↔ $\log_2(1024)$ = 10) **- At least FOUR laws of indices and how they connect** **- At least TWO laws of logarithms and their connection to the index laws** **- At least ONE real-world Kenyan application arrow showing how the model applies** **(b) Write your final, complete answer to the driving question:** **'How can writing numbers as powers and logarithms help us make sense of — and calculate with — the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science?'** **Your answer must:** **- Be at least three paragraphs long** **- Reference the Maize Doubling Puzzle specifically** **- Include at least two other Kenyan real-world examples** **- Explain both the conceptual reason (why it works mathematically) and the practical reason (why it helps people)** **- Show how your understanding has grown across the 8 lessons**	important pieces of information: the base (2, the doubling factor) and the exponent (20, the number of seasons). This compression is not merely cosmetic. The Laws of Indices show that operations on these compressed forms are simpler: multiplying $2^{20} \times 2^5$ requires only adding 20 + 5 = 25 rather than multiplying seven-digit numbers. The difficulty of the original problem collapses into an easy addition. Logarithms extend this power by answering the question indices cannot: given the result, what was the exponent? When a Kisumu fishing cooperative needs to know how many years it will take for a Ksh 50,000 savings deposit to grow to Ksh 200,000 at 8% annual compound interest, they face the equation $(1.08)^n = 4$. The unknown n is trapped in the exponent — a position where ordinary algebra cannot reach it. Applying $\log_{10}$ to both sides and using the Power Law of logarithms brings n down as a coefficient: $n = \log_{10}(4) \div \log_{10}(1.08) \approx 18$ years. The logarithm did not perform magic — it exploited the structural relationship between multiplication and addition that the laws of indices reveal. Similarly, when Rift Valley health workers test the alkalinity of Lake Bogoria's water for flamingo conservation, the pH scale — built on the expression $pH = -\log_{10}[H^+]$ — converts ion concentrations that range across fifteen orders of magnitude into a simple 0–14 scale that any technician can read and compare. The deepest insight of this unit is that indices and logarithms are not two separate topics bolted together — they are one idea, seen from two directions. An index asks: 'Start with the base and exponent; what is the result?' A logarithm asks: 'Start with the base and result; what was the exponent?' Every law of logarithms is a law of indices in disguise. This unity is what makes the tools so powerful across such different fields — from the notebook of a Kitale maize farmer, to the spreadsheet of a Nairobi banker, to the laboratory of a Mombasa food safety scientist. When we write enormous or tiny quantities as powers, we are not hiding complexity — we are revealing the simple, repeated structure underneath it.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding of	Explains ALL laws of indices accurately	Explains most laws of indices correctly	States some laws of indices but relies on

Indices	and with deep reasoning. Unpacks each law using factor expansion or the pattern of repeated multiplication/division. Explains zero and negative indices in terms of the maize puzzle pattern without prompting. No misconceptions present.	and gives reasoning for at least two laws using factor expansion. Minor gaps in explanation of fractional or negative indices but no significant misconceptions.	memorised rules without explanation. Errors or confusion present in negative or fractional index laws. Little or no connection to the pattern of repeated multiplication.
Conceptual Understanding of Logarithms	Defines logarithms precisely as the inverse of indices and articulates the equivalence $\log_a(x)=n \Longleftrightarrow a^n=x$ clearly. States and explains all three logarithm laws and explicitly derives each from the corresponding index law. Correctly applies change-of-base formula with justification.	Defines logarithms correctly and states all three laws. Makes some connection between logarithm laws and index laws. Applies change-of-base formula correctly but may not explain its derivation fully.	Attempts a definition of logarithm but it is incomplete or imprecise. States one or two logarithm laws without connecting them to index laws. Limited or incorrect use of the change-of-base formula.
Evidence-Based Reasoning and Use of the Anchoring Phenomenon	Explicitly references the Maize Doubling Puzzle in multiple sections to motivate and illustrate concepts. Cites at least three specific activities or investigations by name or lesson. Evidence is used to EXPLAIN reasoning, not just assert conclusions. Explanation shows how thinking evolved across the 8 lessons.	References the Maize Doubling Puzzle in at least two sections. Cites at least two specific activities or investigations. Evidence generally supports reasoning, though some connections could be made more explicit.	Mentions the Maize Doubling Puzzle briefly or only once. Cites few or no specific activities. Relies on general statements rather than specific evidence. Limited demonstration that thinking evolved during the unit.
Application to Kenyan Real-World Contexts	Provides detailed, accurate analysis of at least TWO Kenyan real-world contexts. For each context: clearly identifies the quantity, explains why ordinary notation is insufficient, shows how indices or logarithms solve the problem, and interprets the numerical result in plain language. Examples are drawn correctly from farming, finance, engineering, or science.	Provides analysis of at least TWO Kenyan real-world contexts. Most of the four required elements (quantity, limitation, solution, interpretation) are present for each context. Minor gaps in interpretation or explanation of why ordinary notation is insufficient.	Mentions at least one Kenyan real-world context but analysis is superficial. Fewer than three of the four required elements are addressed. Numerical results are stated without interpretation, or examples contain factual errors.
Quality and Completeness of the Revised Model and Final Answer	Revised model clearly shows the three-way relationship (ordinary number ↔ index form ↔ logarithm), at least four index laws, at least two logarithm laws linked to index laws, and at least one Kenyan application. Final answer is three or more well-developed paragraphs that address BOTH conceptual (why it works mathematically) and practical (why it helps people) dimensions, reference the Maize Puzzle and two other Kenyan examples, and demonstrate genuine	Revised model shows the three-way relationship and most required elements; one or two components may be missing or only partially developed. Final answer is at least two paragraphs covering both conceptual and practical dimensions, with references to the Maize Puzzle and at least one other Kenyan example.	Revised model is incomplete or missing key components (e.g. no logarithm laws, no Kenyan application). Final answer is one paragraph or less, addresses only procedural aspects, or makes no clear reference to the Maize Puzzle or Kenyan contexts.

	growth in understanding.		